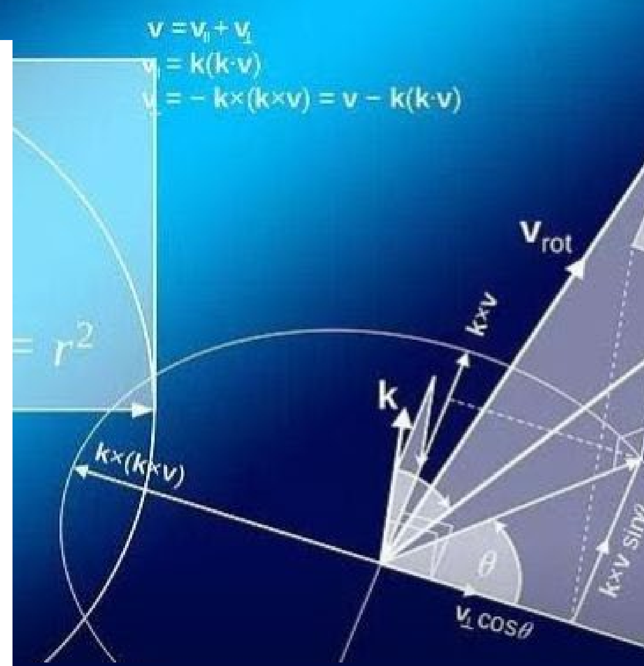

DMISRS Final project M&E report

MARCH | 2024

Compiled by Jessica Horler
(external consultant)

CEA Centre for Educational Assessments

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Introduction

This report summarises the project activities, outputs and participant outcomes of the Diagnostic Mathematics Information for Student Retention and Success (DMISRS) Project between its project lifespan: 2018 - 2024.

The Diagnostic Mathematics Information for Student Retention and Success (DMISRS) Project

The DMISRS Project was developed to address the lack of alignment between students' literacy practices and mathematical proficiency, and the academic practices of their chosen disciplines. This gap results in significant attrition, which continues to widen inequality, and impacts the economy and future development[1]. See Appendix 1 for a summary of 2018 student outcomes for UCT NSFAS recipients, demonstrating a baseline need for the DMISRS project.

The DMISRS project proposal sought to contribute towards the national imperative to address the problem of high failure and drop-out rates, in STEM programmes, specifically in first year Mathematics courses.

The DMISRS Project focused specifically on support for students' mathematical needs in higher education and in STEM programmes in particular, given the national imperative. The project aimed to analyse the NBT Mathematics scores and Higher Education Management Information System (HEMIS) data sets in order to establish how students' performance in the NBTs correlate with graduation and drop-out rates. It also aimed to research the effect of

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[1] Council on Higher Education (CHE), 2020. Briefly Speaking. Extended Programmes with an Integrated Foundation Phase: Theoretical considerations for curriculum design. Number 14. December 2022. Retrieved October 13, 2023, from https://www.che.ac.za/publications/monitoring

Council on Higher Education (CHE), 2021. Vital Stats Public Higher Education 2021. CHE: Pretoria. Retrieved October 13, 2023, from https://www.che.ac.za/publications/vital-stats curriculum-integrated support initiatives for Mathematics in the first year of study and initiate the development of blended learning resources aligned with this research.

Project Evolution

Between its inception and its final year of implementation, the DMISRS project evolved to suit the needs of its participants. This evolution is evident in the project's Theory of Change (ToC) – a roadmap of project outcomes.

At the start of the project, in 2019, the team commissioned an external evaluator to develop a ToC to help guide the project's design, and to serve as a basis for monitoring and evaluation. A ToC describes the causal sequence in which project activities are expected to lead to participant outcomes. A ToC thus helps to ensure project activities are aligned with the goals of the project, and guides the outputs and participant outcomes that should be measured and monitored.

A ToC is not intended to be a static document – it should be critically reflected upon throughout the project lifespan in order to ensure it accurately captures the current iteration of the project. As such, the project ToCs help to illustrate the way in which the DMISRS team used ongoing project data and participant insights to adapt its offering in light of changing needs (particularly in the post-COVID era).

In the image below, the first iteration of the ToC is illustrated. The ToC begins with the project's original primary activities: (1) analyse first-year students' NBT mathematics scores at each of the participating HEIs in relation to their first-year performance, and institutional drop-out and graduation rates; and (2) investigate what curriculum-integrated support programmes are available to these students at each HEI, and what are best practices in this regard. The results of these two investigations are then shared with participating HEIs at annual Teaching and Learning workshops.

As a result of these activities, and direct participation in the annual workshops, HEIs develop an understanding of their students' performance and, consequently, the gaps in their support programmes that fail to address these needs. Using this knowledge, HEIs improve their current student support programmes by using the evidence presented to them in the workshops.

At-risk students are thus better supported by their faculties' revised targeted approach to learning, and, as a result, it is expected that drop-out rates will decrease, and pass/graduation rates will increase.

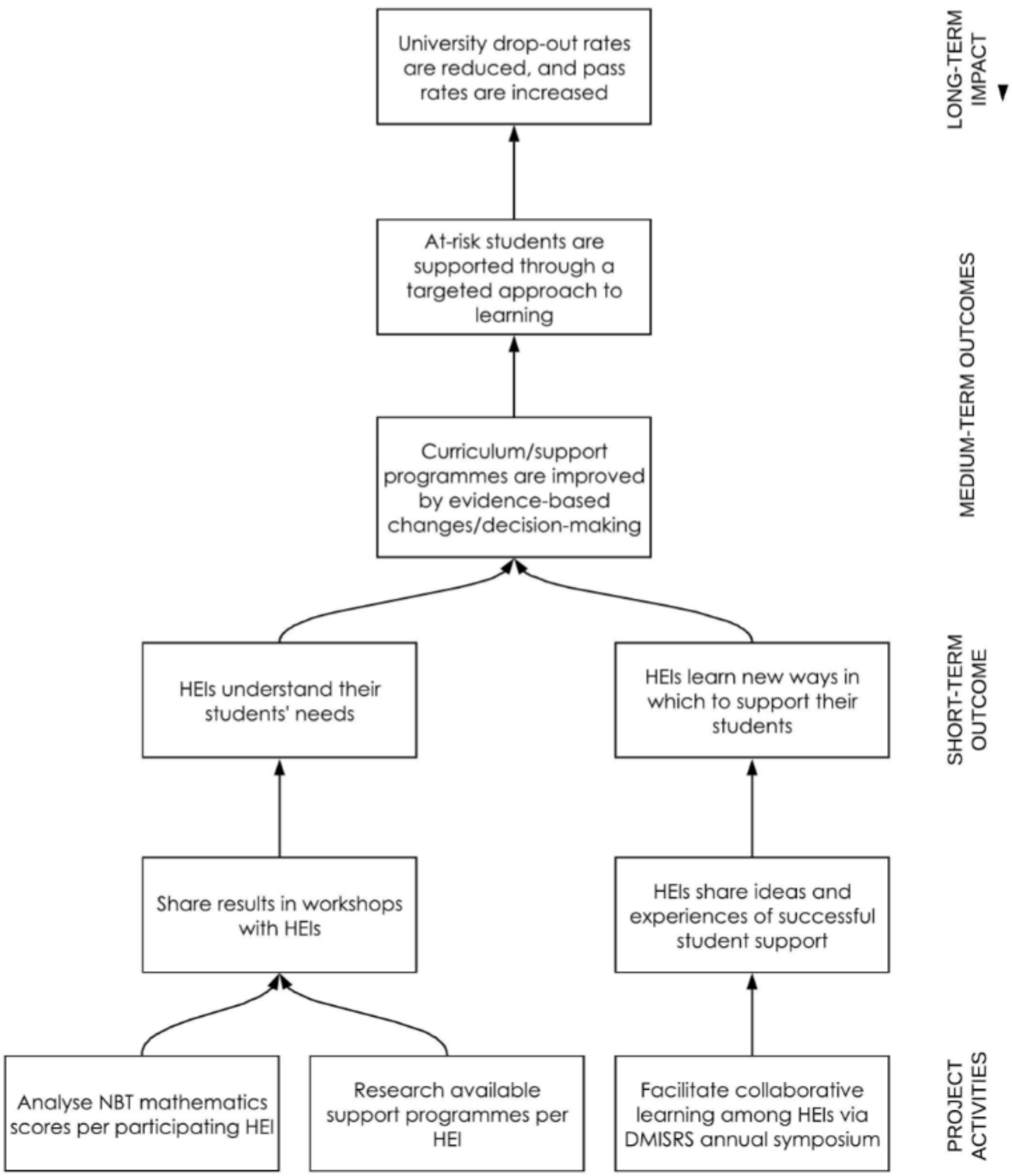


Figure 1: First iteration of ToC

Project Changes

Since first developing the ToC in 2019, the DMISRS team (in collaboration with the M&E consultant) collected event feedback from DMISRS participants after the project engagements. The feedback was used to track participant experiences of the events, as well as to capture suggestions for improving DMISRS offerings. While this feedback was helpful, it was generally not in-depth enough to capture longer-term changes among the participants. For this reason, a qualitative study was conducted at the end of 2021, in which 11 regular attendees at DMISRS events were interviewed by an independent consultant.

In eliciting participant feedback, it is important to note that all survey and interview instruments were kept open-ended. This was an intentional decision due to the nature of the project. As the project is social in nature, it depends on the interaction of multiple people, institutions – each with their own subjective experiences and perspectives. It is likely that participants will benefit from the project in different ways, and using open-ended questions allowed these benefits to be reported on in a more emergent way. As such, the project ToC remains a 'working document' that is likely to change as the project engagements mature and evolve.

Below is a summary of the major themes to emerge from the 2020 and 2021 symposia, as well as the 2021 qualitative study (presented later on in this report).

Table 1. Summary of participant feedback.

2020 Symposium Feedback	**2021 Symposium Feedback**	**2021 Interview Findings**
Two of the biggest takeaways: • The efforts and creativity that academics have made to assist students to transition to an online campus during lockdown. • Most universities faced similar challenges this year.	Two of the biggest takeaways: • The need to embrace technology and innovative teaching and learning methods. • The value of the academic community in terms of sharing common experiences and challenges.	Major findings on project impact: • Collaboration and networking; sharing of ideas, knowledge, common challenges, problem-solving, ideas for resources, etc. • Using diagnostic data

The findings above indicate that the major contribution of the DMISRS project is the community-building that takes place through events and engagements, as well as the use of diagnostic data. This indicated to the DMISRS team that the project's ToC needed to be revised to reflect these participant-driven outcomes.

In addition to the data obtained from the DMISRS participants, the team's experiences of implementing the project highlighted important assumptions and external factors that need to be taken into account when considering the project's impact.

- The team regularly experienced hesitancy among institutional partners to share departmental performance data due to POPIA requirements, as well as general reluctance to reveal poor pass rates. This proved to be a major barrier for the team in their engagements with institutions.
- Turnover among mathematics lecturers often influenced performance data, above and beyond the actual curriculum or student support mechanisms. While the latter two are easier to address, changing a lecturer is not a topic that can be covered by the project. This turnover also has ramifications for the project outcomes – any

changes sparked by the project may be undone by incoming lecturers who have not participated in the DMISRS project.

- The COVID19 pandemic severely hampered the DMISRS project's engagement with participants, watering down any potential for the achievement of outcomes.
- While DMISRS participants may be encouraged to implement new teaching and learning techniques and methodologies, they may not have the power or authority within their department to make changes, especially to the curriculum.

In light of the participant feedback as well as the team's insights, the 2019 ToC was found to be lacking in the following ways:

- It did not include the additional webinars that the team had started to run with an open audience.
- It did not emphasise the value that the project offers participants in terms of community-building (according to the participants' feedback).
- It placed too great an emphasis on student drop-out rates as opposed to institutional student support mechanisms.

In March 2022, the DMISRS team and M&E consultant developed a revised ToC to reflect the qualitative evidence described above. It is presented overleaf, and can be described as follows:

The DMISRS team analysed NBT scores for higher education institutions (HEIs) participating in the project, and also researched the current student support programmes available at each HEI. Using this data, the team engaged in three major activities: (1) this data was shared in institutional workshops; (2) the team ran webinars on the use of diagnostic data; and (3) annual symposia were held every year to facilitate collaborative learning.

The resultant outcomes chain was conceptualised as follows:
As a result of the institutional workshops, HEIs will understand their students' profiles, and associated needs, and use the space to reflect on their teaching and learning practices. There will also be an improved understanding of the use of diagnostic data, as well as *how* to analyse this data. The latter outcome is also realised through the periodic webinars run by the DMISRS team. In addition, these webinars enable HEIs to share commonly experienced

challenges, without clear-cut solutions. This is also encouraged by the annual DMISRS symposia. The symposia also catalyse increased inter-institutional interactions and relationships, and provide a collaborative space for HEIs to share their teaching and learning experiences, as well as their ideas and experiences of successful student support.

All of the aforementioned short-term outcomes are expected to help HEIs learn new teaching and learning strategies, learn new ways to support their students and enable academics to feel a sense of belonging to a community. In the long term, this may help to ensure that curricula and support programmes are improved collaboratively by evidence-based decision-making, and reflecting on both diagnostic data and empirical theory (all of which are encouraged through the DMISRS activities). This contributes to the ultimate impact of the DMISRS project: at-risk first-year mathematics students are supported through a targeted approach to learning.

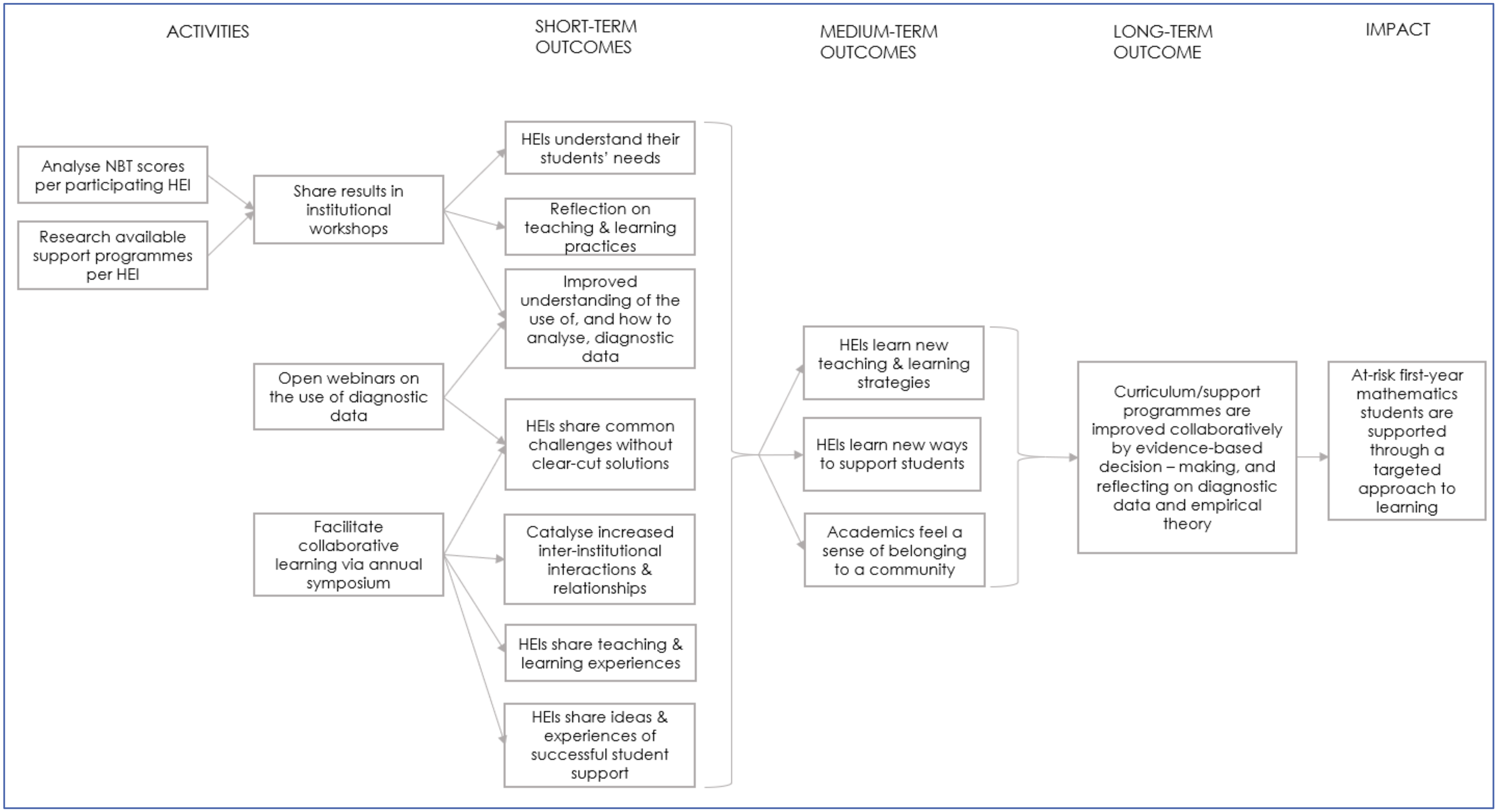


Figure 2: Revised ToC: March 2022

Finally, in 2023, the DMISRS team recognised a need to cater for individual students' mathematical profiles through adaptive learning solutions, in response to NBT diagnostic reports. This was addressed by designing a number of pilot projects to craft a solution to student support and use early indicators to understand students' engagement, experience and impact on learning. The final ToC is illustrated below, with this additional mechanism for student support (highlighted in blue).

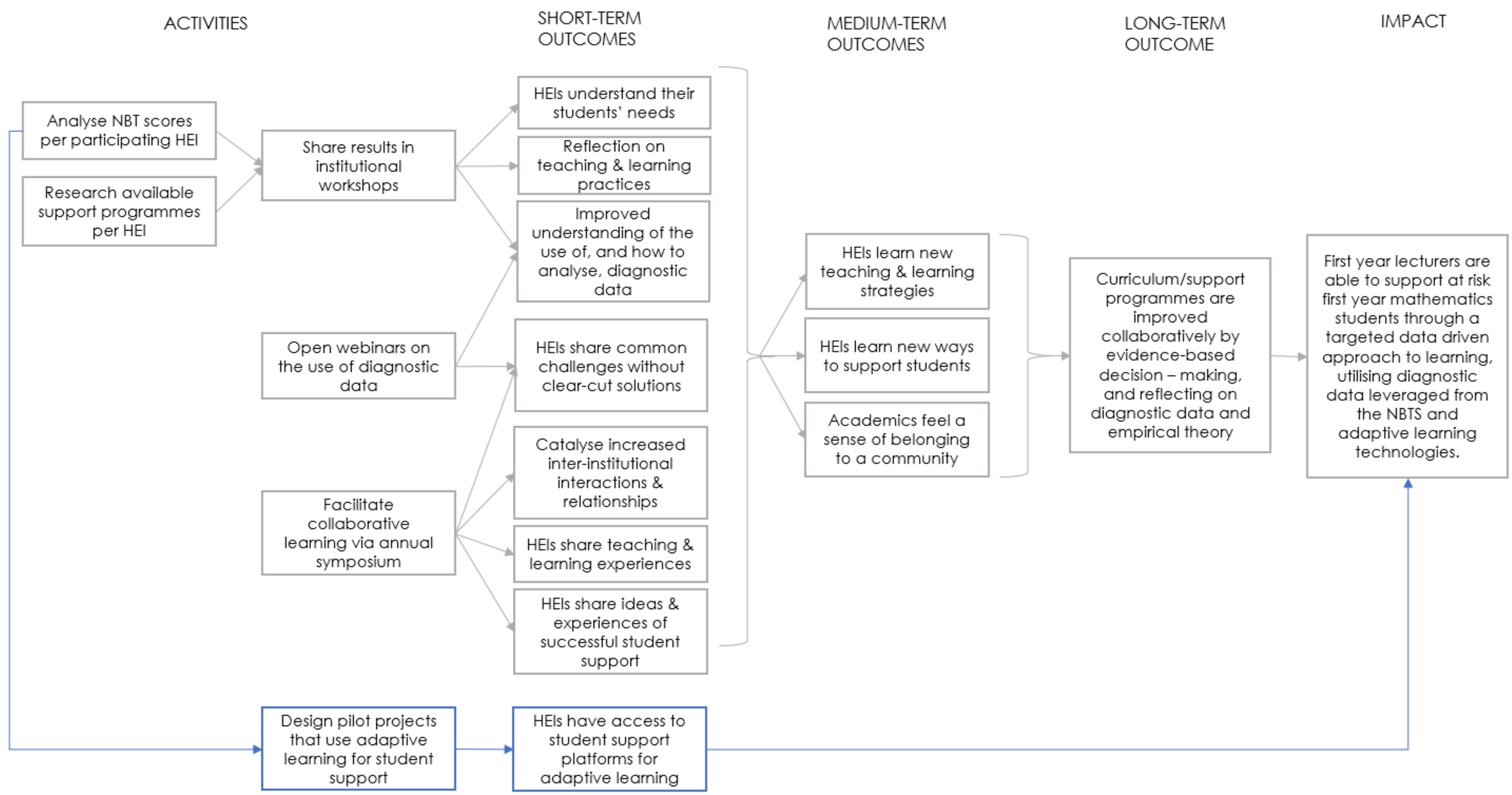


Figure 3: Final ToC: 2023

DMISRS Project Activities

In the following sections, a retrospective summary of project activities for the DMISRS project is provided.

HEI Support

Student Support Surveys

From 2018 to 2021, the DMISRS team surveyed project participants regarding the student support mechanisms that are available to students, and to understand how HEIs used the NBTs.

Table 2. Summary of student support surveys.

Year	Number of HEI respondents	Findings Summary
2018	5	The majority of universities (86%) do not use the NBT results for student placement, or to identify support needs.
2019	16	• The majority of universities (88%) do not use the NBT results for student placement, or to identify support needs (75%). • Half of the respondents reported that their student support services could be improved. • Fifty-four percent of respondents said that they need support in order to change their student support services.
2020	13	• The majority of universities (88%) do not use the NBT results for student placement, or to identify support needs (75%). • 75% of respondents reported that their student support services could be improved. • 69% of respondents said that they need support in order to change their student support services.

2021	There were not enough survey respondents to complete a meaningful analysis. This reflected a trend of survey fatigue among project participants during the COVID-19 pandemic.

Teaching and Learning Workshops

In 2019, the DMISRS team initiated customised institutional workshops with HEI partners. The workshops provided an overview of the DMISRS project, as well as the academic literacy (AL), quantitative literacy (QL) and MAT test domains and how they are assessed in the NBTs. Workshop objectives included:

1. Discussing the relevance of the subdomains to teaching and learning;
2. Providing feedback on student performance (broken down into a national profile, an institutional profile, and departmental cohort); and
3. Facilitating a discussion about the resultant implications for teaching and learning at each institution.

Unfortunately, these in-person engagements were curtailed by the COVID-19 pandemic from 2020 onward, as the project refocused its activities in light of the changing needs within higher education. They were re-initiated in 2023 by the DMISRS team, focusing on analyses of institutional data, as well as the current initiatives underway to utilize the diagnostic data to bridge the articulation gap between school and higher education using adaptive computer aided learning.

Table 3. Summary of teaching and learning workshops.

Year	Number of Workshops	HEIs	Feedback
2019	4	Rhodes University (RU), Mangosuthu University of Technology (MUT), University of Zululand (UNIZULU), and Safako Makgatho Health Sciences University (SMU)	• 99% of all workshop participants found a particular discussion or presentation interesting or useful. • 93% of participants intended to apply lessons they learned in the

			workshop in their future planning. • 64% of participants felt as though something was missing from the workshop.
2023	5	WITS University, University of the Free State, Rhodes University, Nelson Mandela University, University of Zululand	Feedback data not collected

Collaborative Learning

Webinars

Due to the pandemic, the DMISRS team shifted away from in-person institutional workshops to online webinars with a wider audience. As with the Teaching and Learning workshops, the objectives of these webinars were to discuss the use of student diagnostic data for wrap-around student support and curriculum development.

Table 4. Summary of webinars.

Year	**Format**	**Number of attendees**	**Feedback**
2020	Online	34	Only six attendees opted to provide feedback. All six respondents found a particular presentation interesting or useful; half noted that the presentation on ongoing work with partner institutions to be the most interesting/useful. The majority of respondents (83%) said that they feel as though they have a better understanding of how they would go about determining their students' needs, as well as supporting these needs. Closing comments were positive, thanking the facilitators for an interesting

			discussion and for fostering a feeling of community.
2020	Online	101	All but one respondent finding a discussion/presentation particularly interesting or useful. A quarter of respondents reported they were particularly interested in the usefulness of using diagnostic assessment to inform curriculum development. Another quarter specifically mentioned the analysis of NBT sub-domains and its usefulness in providing more tailored, student support. Almost all (94%) of the respondents reported that they intend to apply what they felt was their biggest takeaway to their future planning, such as developing diagnostic measures and tools to use in formative assessments and curriculum development.
2022	Online	101	Too few attendees opted to provide feedback, prohibiting meaningful analysis.

DMISRS Symposia

Since its inception, the DMISRS team has arranged and facilitated annual symposia in which mathematics academics from around the country came together to share best practices and new research in teaching and learning.

Table 5. Summary of symposia.

Year	Theme	Format	Feedback
2018	Setting the scene, engaging participation and negotiating collaboration	In-person	• Feedback was provided in plenary and through discussion. The following themes arose from these discussions: o How do curricula compare across institutions o Creation or strengthen CoPs, discussion forums

			o The need to better understand student challenges, diversity and motivation o Identifying common challenges o The extent of the use of blended learning resources and what constitutes best practice.
2019	Mathematics in Practice	In-person	• All participants said that they found a discussion/presentation particularly interesting or useful. • The majority of participants (89%) reported that they intend to use what they had learned at the symposium in their future planning.
2020	*First year mathematics solutions: the use of assessments, performance, blended learning and curriculum development in the time of covid-19.*	Online	• Regarding the effectiveness of the novel online format, the majority of survey respondents rated it as *very good* (69%) or *excellent* (15%). • All but two participants said that they found a discussion/presentation particularly interesting or useful. • The majority of participants (80%) reported that they intend to incorporate what they had learned at the symposium into their future planning.
2021	*First year mathematics: from crises mode to the new 'normal'*	Online	• Regarding the effectiveness of the novel online format, the majority of survey respondents rated it as *very good* (30%) or *excellent* (55%). All participants said that they found a discussion/presentation particularly interesting or useful.

			• The majority of participants (75%) reported that they intend to incorporate what they had learned at the symposium into their future planning.
2022	*What does it take to teach Mathematics for and in higher education?*	Online	Regarding the effectiveness of the hybrid format, the majority of survey respondents rated it highly, scoring it 5 (79%) out of 5. When asked whether there was any discussion or presentation that was particularly interesting or useful, 100% of respondents said that there was. All respondents said that they intend to apply this lesson to their future planning
2023	Lessons learnt: Diagnostic Mathematics Information for Student Retention and Success (DMISRS) case studies in teaching first-year mathematics in South Africa	In-person	Feedback was collected as part of the writing workshop that constituted the last session of the symposium. The synthesis of these discussions is in process.

Figure 4: Word cloud depicting respondents' biggest takeaways from the 2021 symposium

Figure 5: Word cloud depicting respondents' biggest takeaways from the 2022 symposium

Conference Presentations

In order to share the findings of research activities, the DMISRS team presented and participated in a number of national conferences.

Table 6. Summary of conference presentations.

Year	Conference	Participation
2023	Association for Mathematics Education of South Africa (AMESA)	• Tatiana Sango: “Understanding the motivation factors of prospective students in the STEM disciplines”. • Mathematics school educator workshops were prepared for presentation to school educators by Sanet Steyn, titled: “NBT Mathematics and quantitative literacy: exploring the connections and gaining insights”.
2023	FREMO	• Tatiana Sango and Sanet Steyn presented: “Mathematics and Academic literacy cross-domain analysis in high-stakes assessment for augmented practices in diagnostic interpretation.”
2023	DELTA	• Tatiana Sango and Sanet Steyn presented: “Cross-domain mapping in diagnostic assessment: enhancing interpretation for student support.”
2023	PME-45	• Anthony Essien presented: “Using student-generated examples to understand preservice teachers’ preparedness to teach mathematics.”
2022	SALALS	• Sanet Steyn and Tatiana Sango presented: "Academic literacy demands in Mathematics problem-solving: Polya's strategy through the AL lens"
2022	SALALS	• Robert Prince presented: "Does Higher Education entrance assessments in Academic and Quantitative Literacies predict success in STEM programmes?"
2022	IACAT	• Tatiana Sango & Precious Mudavanhu presented: "Investigating feasibility of Computerized Adaptive Testing (CAT) for NBT Mathematics assessment."
2022	IEEE Procom conference	• Robert Prince and xxx presented a joint paper: “Does Higher Education Entrance Assessment in Academic Literacy Predict Success in Engineering Study?”

2021	European Society for Engineering Education	• Robert Prince and Simpson presented: "Do Higher Education Entrance Exams Predict Success in Engineering Study?"
2019	REES	Robert Prince, Mutakwa and Dunlop presented: "The diagnostic potential of admissions tests for South African Higer education."
2019	A-MoDe Conference	• Robert Prince and Jacob Jaftha presented: "Teaching, learning and assessment of multimodal digital academic numeracy practices."
2019	SAARMSTE	• Benita Nel attended and presented at SAARMSTE on the 15 – 17 January 2019.
2018	HELTASA	• Anita Campbell presented: "Developing an understanding of what the role of mathematics lecturer should be".
2018	AMESA WC	• Anita Campbell presented: "Get to know your students through the NBT data."

Student Support

First-year Student Support

In 2023, to assist HEI partners with blended learning, the DMISRS team initiated a pilot involving digital solutions for mathematics. This is a student success initiative, which involved a cohort of approximately 100 first-year students and 100 NBT MAT writers (applicants to university), to understand the extent to which NBT data on sub-domain importance, coupled with the use of an adaptive learning platform (such as ALEKS), can improve student outcomes.

Mathematics Adaptive Learning: ALEKS

ALEKS is a digital solution system founded by Dr Jean-Claude Falmagne (New York University and the University of California, Irvine) and owned by McGraw Hill. The artificially intelligent learning and assessment system utilises big data and machine learning. Using ALEKS and NBT data, the DMISRS team established a student support proof-of-concept to support first-year students demonstrating gaps in mathematics knowledge. The aim of this pilot was to investigate the potential of using an adaptive learning system to facilitate a smoother transition for students entering tertiary education.

The NBT ALEKS solution uses the NBT data and relative importance analysis to inform course-specific parallel intervention programmes that do not require additional time from lecturers. It also integrates with the course and supports students' concept development by closing the gaps in their prior learning – which has been identified by the NBT diagnostics – while having most impact in the courses they are enrolled in.

In June 2023, the digital solution was first piloted in four first-year mathematics courses among different student performance categories: those who failed the first-semester course but qualified for supplementary examination and students who passed the first-semester course, but whose final course mark was low. It was then piloted again in September 2023 with Grade 12 learners who were part of the UCT 100UP Project[2] and transitioning into their first university year in 2024. A final pilot took place in collaboration with the Faculty of Health Sciences at UCT, looked at foundational mathematical skills. A small study looked at creating a course for psychology students with a subset of 47 different skills in the intervention programme.

While the pilot data is still being analysed, preliminary analyses indicated that there is a correlation between student engagement and the improvement in performance.

Prospective Student Support (Grade 11)

The implementation of Computerised Adaptive Testing (CAT) technology into assessment for an early academic profiling and preparedness for higher education may enable the early identification of students' academic strengths and weaknesses, providing valuable insights into the development of critical skills and allowing for targeted interventions and support as

[2] 100UP is a UCT initiative addressing the problem of demographic under-representation in higher education by targeting school learners from disadvantaged backgrounds and coaching them towards university access.

needed. The work is intended to contribute to enhancement of the overall quality of pre-university education and improving student readiness for higher education.

Mathematics Adaptive Testing: CAT

The DMISRS team piloted Computerised Adaptive Testing (CAT) assessment approach to investigate the key trajectories of academic preparedness among high school learners. The Grade 11 Mathematics assessment instrument was designed with a specific focus on understanding how CAT leverages adaptive technology to customise the difficulty of test items based on each student's demonstrated proficiency level. This approach ensures that the assessment provides accurate and actionable insights into students' mathematical abilities.

Analyses of the pilot data (obtained from the administration of the CAT assessment to 77 Grade 11 learners in Khayelitsha school) will help with the development of a solution for early diagnostics and support initiatives at Grade 11 level. The pilot indicators are promising and when data is put in conversation with educators, it is evident that the targeted intervention during the final schooling year could make a difference in learners' development of mathematical skills requiring attention (e.g., MAT M5 – Geometrical reasoning subdomain in NBT MAT as an example in Figure 6).

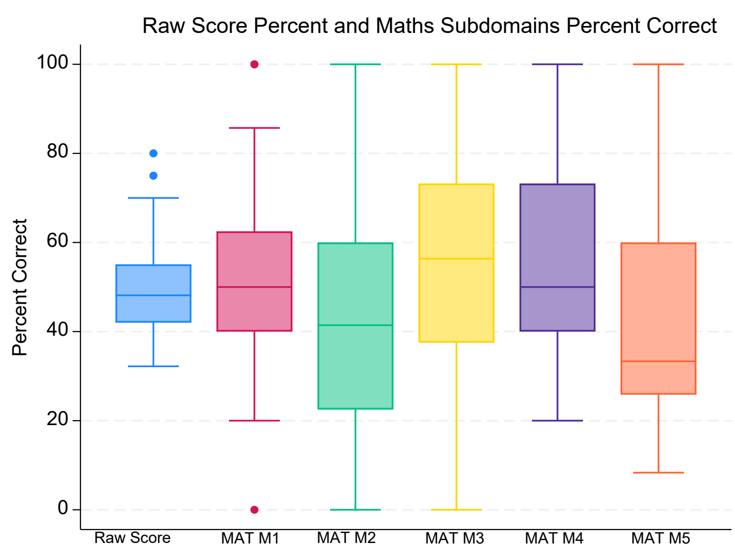


Figure 6: CAT Mathematics performance: Grade 11 Pilot 2023

Amathuba Pilot

The Mathematics Amathuba Pilot sought to monitor and assess student support initiatives utilising the Amathuba learning management system, with impact assessed from the data metrics generated from the platform.

The NBT MAT self-paced course for Mathematics online candidates is structured around the student experience with the online test and includes information about the test, technical set up, test requirements, sample test items and quizzes.

In the first administration of the pilot, approximately 102 active users took part, which increased to approximately 154 in the second administration.

Feedback data was collected from students, which was positive overall. The majority of students rated the overall organisation and structure of the course highly, as well as its coverage of the steps involved in navigating the online NBT platform. More than 90% of students found the learning materials relevant and helpful. Of the students who provided feedback, 67% reported feeling confident in taking the NBT online after completing the course.

DMISRS Project Impact

2021 Qualitative Study

Key outcomes of the DMISRS project (see the Theory of Change) are to help participating institutions understand their students' needs and to learn new ways in which their first-year mathematics students can be supported. These topics form the basis for each year's symposium, as well as the institutional workshops held with participating HEIs.
While these outcomes are regularly probed with feedback surveys following these events, it was deemed necessary to plan a more in-depth investigation into participants' views of the project, especially in light of the survey fatigue noted earlier.

In November 2021, eleven DMISRS participants were interviewed by an external consultant in order to understand how participants have engaged with the project over the years, what impact it has had on their work, and what they would like to see from the project going forward. A full report was compiled by the M&E team and is available upon request.

The interviews highlighted the positive impact that the project has had on interviewees' teaching practices, primarily through its collaborative nature. Sharing resources, perspectives, experiences and common challenges have been a source of growth and comfort to project participants, especially as higher education has moved online.

"I think that the symposia as well as the workshops were useful to me in the sense that they gave me a better picture of what others experience; sometimes there are differences and sometimes there are similarities…It's useful to hear other people's voices and experiences."

While measuring the impact of the DMISRS Project is difficult in such fluid and emergent conditions, and as many events have had to be cancelled or changed due to the pandemic, a definite sense of learning and positive growth were noted among the interviewees. Furthermore, every interviewee had a compliment to bestow on the DMISRS team and the project in general, indicating widespread appreciation for the project and the lengths that the team are willing to go to in order to support the members.

"It's been good, supportive...it's motivational that the project exists. It's always done with a very good sense of care, good follow-up, calls for papers, it's well organised. It's a nice comforting space."

Conclusions

Despite contextual and internal challenges, the DMISRS team consistently met its project objectives year-to-year, and implemented a number of contextually relevant interventions among its HEI community. Partners consistently praised the collective learning opportunities facilitated by the project.

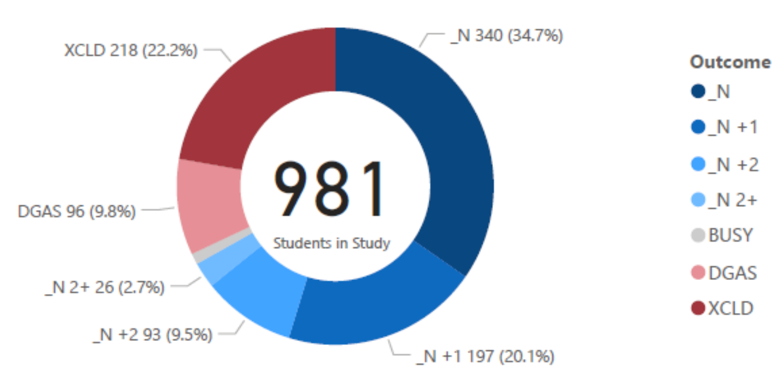
The project team also piloted several blended learning interventions in light of the ongoing need from HEI partners to incorporate technology into the classroom, and to assist with the preparation of first-year mathematics students.

Key learnings from the project include:

- The need in the tertiary education sector for collaborative learning and peer support among academics.
- Significant gaps in the high school system in terms of preparing students for university-level mathematics, particularly following the COVID-19 pandemic.

Appendix 1: 2018 NSFAS Student Outcomes

2018 Cohort Outcome



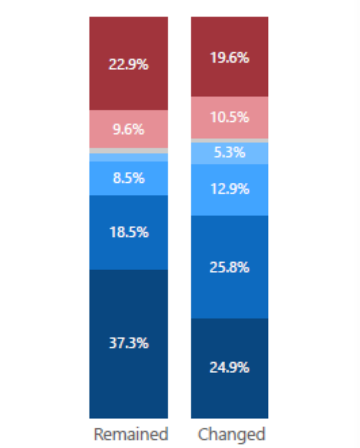
Average NBT by Outcome

Outcome	Count	nbt_al	nbt_ql	nbt_math
_N	340	63.07	52.42	63.46
_N +1	197	60.22	52.30	70.39
_N +2	93	61.39	56.68	75.28
_N 2+	26	61.46	57.96	75.82
BUSY	11	59.73	53.36	78.18
DGAS	96	60.24	51.65	68.60
XCLD	218	58.13	53.12	75.20
Total	**981**	**60.94**	**53.03**	**69.78**

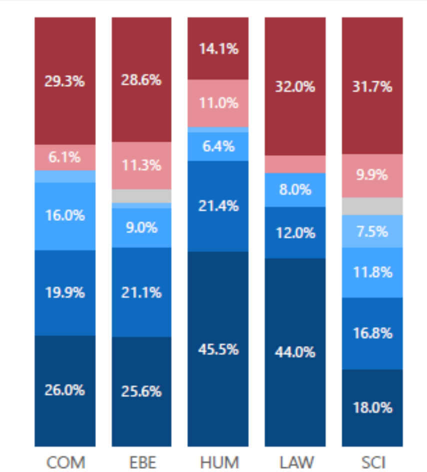
Reference

Outcome	Description
_N	Graduated in minimum time
_N +1	Graduated in minimum time + 1 year
_N +2	Graduated in minimum time + 2 years
_N 2+	Graduated in more than minimum time + 2 years
BUSY	Pending
DGAS	Dropped Out in Good Academic Standing
XCLD	Excluded

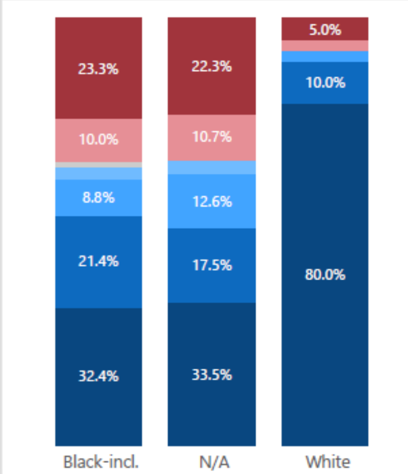
Outcome by whether they stayed in their original program or not



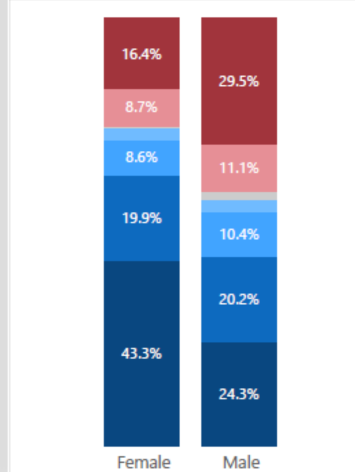
Outcome by Faculty



Outcome by Population Group



Outcome by Gender



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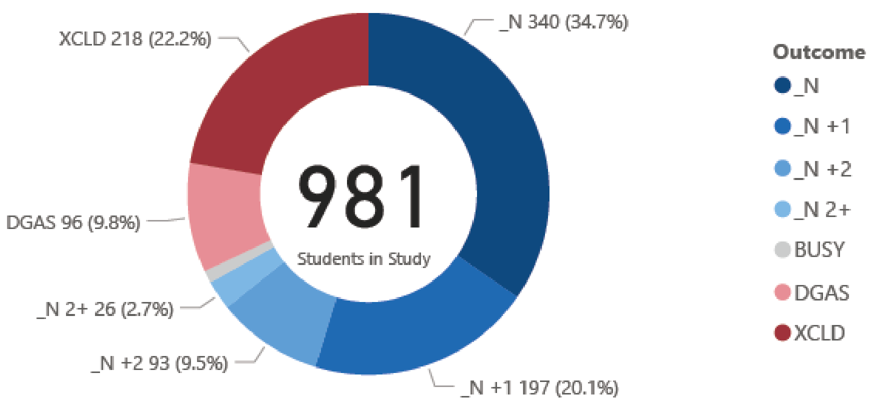

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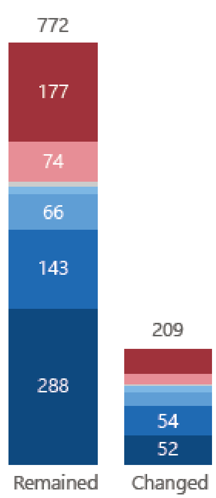
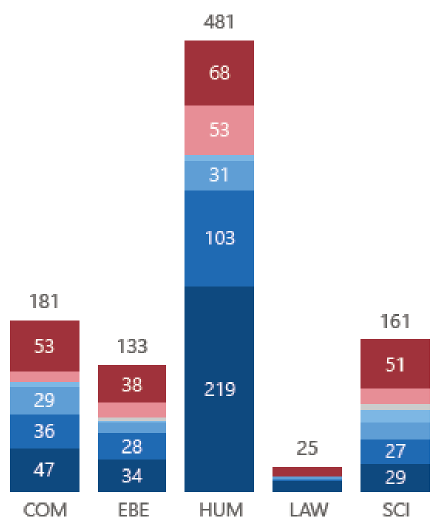
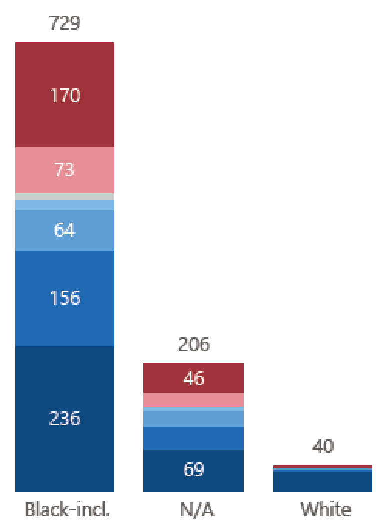
Outcome by whether they stayed in their original program or not



Outcome by Faculty



Outcome by Population Group



Outcome by Gender